

INTERMEDIATE

DAX & Power Query Cheat Sheet

Once you're comfortable with Excel formulas, this is the bridge into Power BI: transforming data with Power Query, then measuring it with DAX.

Part 1 — Power Query (Data Transformation)

Power Query is where you clean and reshape data *before* it hits your model. Access it via Home → Transform Data.

Function / Step	What it does	Example
Remove Duplicates	Drops duplicate rows based on selected columns	Home → Remove Rows → Remove Duplicates
Split Column	Breaks one column into several (by delimiter/position)	Transform → Split Column → By Delimiter
Unpivot Columns	Turns wide data into long/tall format for analysis	Transform → Unpivot Columns
Merge Queries	Joins two tables on a matching column (like VLOOKUP)	Home → Merge Queries
Append Queries	Stacks two tables with the same columns on top of each other	Home → Append Queries
Group By	Aggregates rows (sum/count/avg) by a category	Transform → Group By
Fill Down	Copies a value down to fill blank cells below it	Transform → Fill → Down

Part 2 — DAX Fundamentals

DAX (Data Analysis Expressions) is the formula language behind Power BI measures and calculated columns.

Function / Step	What it does	Example
SUM / AVERAGE	Basic aggregation over a column	Total Sales = SUM(Sales[Amount])
CALCULATE	Changes the filter context of a calculation — the single most important DAX function	Sales_WestOnly = CALCULATE([Total Sales], Region[Area]="West")
SUMX	Row-by-row calculation, then sums the result	Revenue = SUMX(Sales, Sales[Qty]*Sales[Price])
RELATED	Pulls a value from a related table (across a relationship)	Category = RELATED(Products[Category])
FILTER	Returns a filtered table, usually used inside CALCULATE	CALCULATE([Total Sales], FILTER(Sales, Sales[Qty]>10))
ALL	Removes filters — used for % of total calculations	PctOfTotal = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Sales)))

Function / Step	What it does	Example
DIVIDE	Safe division that handles divide-by-zero automatically	Margin = DIVIDE([Profit], [Revenue], 0)

Time Intelligence (the DAX functions that trip people up)

Function / Step	What it does	Example
TOTALYTD	Year-to-date running total	YTD Sales = TOTALYTD([Total Sales], Calendar[Date])
SAMEPERIODLASTYEAR	Compares to the same period one year back	LY Sales = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Calendar[Date]))
DATEADD	Shifts a date column by a number of periods	PrevMonth = CALCULATE([Total Sales], DATEADD(Calendar[Date], -1, MONTH))

Time intelligence functions require a proper Date/Calendar table marked as a Date Table in the model — this is the #1 cause of them silently returning blank results.

Measure vs. Calculated Column — the rule of thumb

	Calculated Column	Measure
Computed	Once, per row, stored in the model	On the fly, whenever it's used in a visual
Use for	Something you'll slice/filter by (e.g. a category)	Aggregations you'll display (e.g. Total Sales, Margin %)
Cost	Increases file size	No storage cost, slightly more calculation at query time

Ready to go deeper? The Statistical Analysis Cheat Sheet (Advanced) covers the stats concepts that turn dashboards into genuine insight.
Terence Rebello · The Analyst's Toolbox · terencerebello.com